

Talk 5. Vector bundles on the Fargues - Fontaine Curve I.

Motivation: isocrystals

Def: Given (A, φ) w/ A ring & $\varphi \in \text{End}(A)$. An isocrystal is a $\overset{\text{fin. proj.}}{A}$ -mod. M w/ iso. $\varphi_M: \varphi^* M \xrightarrow{\sim} M$

e.g. $A = W(\overline{\mathbb{F}}_p)[1/p] = \check{W}_p$ w/ $\varphi = \varphi_{\overline{\mathbb{F}}_p} \stackrel{= \text{Frob}}{=} \varphi_{\overline{\mathbb{F}}_p}$. $\varphi\text{-mod.} / (A, \varphi) =: \overline{\mathbb{F}}_p\text{-isocrys.}$

1) 1-dim. case: $M = A$, $\varphi_M: 1 \mapsto p^m$ for $m \in \mathbb{Z}$

2) For $\lambda = d/r \in \mathbb{Q}$, $r > 0$, d, r coprime. Then

$$D(\lambda) = \check{W}_p^{\oplus r}, \varphi_\lambda \leftrightarrow \begin{pmatrix} 1 & & & 0 \\ & \ddots & & \\ & & p^d & \\ 0 & & & 1 \end{pmatrix}$$

Fact: The $D(\lambda)$ are all simple.

Thm (Dieudonné - Manin): The cat. of $\overline{\mathbb{F}}_p$ -isocrys. is semi-simple w/ simple obj $D(\lambda)$.

- These show up as rat'l crystalline coh. gps. e.g. $A/\overline{\mathbb{F}}_p$ ab. var., this is the "rat'l Tate module for $\ell = p$ "

- These are the easiest part of the "semi-linear" machinery in p -adic Hodge theory.

- FF curve: It will "linearize" these (i.e., from s.e. alg. to l.a.)

Back to X_{FF} : $\overset{\text{alg. cl.}}{\vee}$

Fix C^b perfectoid in char p . (we fix our base $\check{W}_p$), ω pseudo-uniformizer

Fact: $X_{\text{FF}}^{\text{ad}} \simeq \mathcal{Y}/\varphi^{\mathbb{Z}}$, $\mathcal{Y} = \text{Spa}(A_{\text{mf}}) \setminus \{p[\omega]\}$
 $\mathcal{O}_{\mathcal{Y}}(\mathcal{Y}) = B$

Recall: We constructed B as "holo. fns" in the variable p on the punctured unit disc.

$\uparrow$ this space is $\mathcal{Y}$

These motivates X_{FF} :

1) People knew $\left\{ \begin{array}{l} \text{cat. of } p\text{-adic Hodge} \\ \text{theoretic objs} \end{array} \right\} \simeq \left\{ \begin{array}{l} \text{vbs on } \mathbb{D}^{\times} + \text{Frob.} \\ \text{s.h. data extra stuff} \end{array} \right\}$

see e.g. Brinon - Conrad

①

② Want to quotient by φ . φ is not an automorphism on $\mathbb{A}^X$

Let's construct vb. on X_{FF} from isocrystals!

We have $y \rightarrow \text{Spa } \mathbb{A}_p \Rightarrow$ isocrystals can be pulled back to y + descended to X_{FF}^{ad} ! In particular, $\mathbb{Z} \hookrightarrow \text{Pic}(X_{FF}^{\text{ad}})$.

Goal: $\{ \text{vbs on } X_{FF} \} / \sim \leftrightarrow \{ \text{isocrys. / } \mathbb{F}_p \} / \sim$ as a set, not equiv of cats.

Recall: $X_{FF} = \text{Proj}(P)$, $P := \bigoplus_{n \geq 0} B^{\varphi=p^n}$

First: $\text{Pic}(X_{FF}) \simeq \mathbb{Z}$

Key property: P is graded factorial w/ fixed. elts in deg 1 (& these vanish w/ simple zero at one pt.) $(\simeq \mathbb{P}_{\mathbb{C}}^1 :)$

For any $m \in \mathbb{Z}$, $\exists \mathcal{O}(m) \in \text{Pic}(X_{FF})$, def. by

$$\mathcal{O}(m) := \bigoplus_{n \geq m} B^{\varphi=p^n} = \bigoplus_{n \geq 0} (B \otimes D(-m))^{\varphi=p^n}$$

Note that $B^{\varphi=p^n} \simeq (B \otimes_{\mathbb{W}_p} D(-n))^{\varphi \otimes \varphi_n = \text{id}}$

(works for any $D(\lambda)$) $\triangle$

Prp : $\text{Div}(X_{FF}) \xrightarrow{\sigma} \text{Pic}(X_{FF})$ factors through $\mathbb{Z}$
 $\downarrow \text{deg} \quad \nearrow \mathcal{O}(n)$
 $\mathbb{Z} \xrightarrow{n}$

Pf: Use key property to show that $x \in X_{FF}$, $\mathcal{O}(x) \simeq \mathcal{O}(1)$. $\square$

Prp : $B^{\varphi=p^n} \xrightarrow{\sim} H^0(X_{FF}, \mathcal{O}(n))$.

Pf: next time. $\square$

Cor: $\text{Pic}(X_{FF}) \simeq \mathbb{Z}$.

Now, let's construct $\mathcal{O}(\lambda)$ for $\lambda \in \mathbb{Q}$ ($\lambda = d/r$, $r > 0$). (Rmk: $\mathcal{O}(\lambda) \leftrightarrow D(-\lambda)$).

- For any $E/\mathbb{W}_p$ fin. $\uparrow$ unr. have $X_{FF,E}$: key property also works for $X_{FF,E}$

$$X_{FF,E} \simeq X_{FF} \otimes_{\mathbb{W}_p} E$$

$\downarrow$ fin. ét. of deg

$X_{FF} [E: \mathbb{W}_p]$

$$\downarrow$$

$$\text{Pic}(X_{FF,E}) \simeq \mathbb{Z}$$

②

Fact: $E/\mathbb{Q}_p$ unr. then $X_{FF, E}^{\deg h} \cong X_{FF} \otimes_{\mathbb{Q}_p} E \cong \text{Proj} (P \otimes_{\mathbb{Q}_p} E)$

Pf: It will follow from: $B^{\ell} = P^n \otimes_{\mathbb{Q}_p} E \cong B^{\ell^h} = P^{nh}$ (pick $\langle \gamma \rangle = \text{Gal}(E/\mathbb{Q}_p)$ - action on RHS by γ acting via $P^{-n} \ell$) $\leadsto (B^{\ell^h} = P^{nh}, \text{Gal}(E/\mathbb{Q}_p) \cong B^{\ell} = P^n)$ $\square$.

Let E_h be the unr. extn. of $\mathbb{Q}_p$ of deg h .

Def: $\mathcal{O}(\lambda) := \pi_{r,*}(\mathcal{O}_E(d))$ It is a rk r v.b.

Rmk: In particular, $\pi_r^* \mathcal{O}(\lambda) \cong \pi_r^* \pi_{r,*}(\mathcal{O}_E(d)) \cong \mathcal{O}_E(d)^{\oplus r}$ ($E_r \otimes_{\mathbb{Q}_p} E_r \cong E_r^{\oplus 1}$)

Def: $Z \in \text{Sch}$ is a "complete alg curve":

- 1) Z is conn. sep., Noeth & reg. of dim 1 divisors
- 2) $\text{Pic}(Z)$ admits a deg morphism taking positive values on effective $\neq 0$ Weir

Def: Fix $\mathcal{F}$ a vb on Z . Then

- 1) $\deg(\mathcal{F}) := \deg(\wedge^{\text{rk}(\mathcal{F})} \mathcal{F})$ 2) For $\mathcal{F} \neq 0$, the slope $\mu(\mathcal{F}) := \deg(\mathcal{F}) / \text{rk}(\mathcal{F})$

have: $\mu(\mathcal{F} \otimes \mathcal{G}) = \mu(\mathcal{F}) + \mu(\mathcal{G})$

Prop: Given $u: \mathcal{F} \rightarrow \mathcal{G}$ map of vbs s.t. u is generically an iso. then $\deg(\mathcal{F}) \leq \deg(\mathcal{G})$

w/ " $=$ " $\Leftrightarrow u$ is an iso.

Def: $\mathcal{F} \neq 0$ v.b.

- i) $\mathcal{F}$ is semistab. if $\mu(\mathcal{G}) \leq \mu(\mathcal{F})$. $\forall \mathcal{G} \subset \mathcal{F}$ non-zero subbde.
- ii) $\mathcal{F}$ is stab if $- " - < - " - \forall \mathcal{G} \subsetneq \mathcal{F} - " -$.

e.g. "All line bdes are stab"

- 2) For isocrystals, the stab objs are the $D(\lambda)$ semi-stab $D(\lambda)^{\oplus r}$

Prop: Each vb. $\mathcal{F}$ on Z has a unique functorial "Harder-Narasimham" filn:

$$0 = \mathcal{F}_0 \subset \mathcal{F}_1 \subset \dots \subset \mathcal{F}_r = \mathcal{F}$$

s.t. $\mathcal{F}_i$ are vbs & $\mathcal{F}_i / \mathcal{F}_{i-1}$ is semi-stab. s.t. $\mu(\mathcal{F}_r / \mathcal{F}_{r-1}) < \dots < \mu(\mathcal{F}_1)$

Pf: For $\mathcal{F}_1$, pick max. rk & deg. $\square$

③

Prop: For $\lambda \in \mathbb{Q}$, the subcat. $\mathcal{C}_\lambda^{\text{sst}}$ (vbs semistab. of slope λ) is ab. & of fin length, i.e., each obj has a fin filn w/ simple graded quotients

Pf: Any proper sub-obj must have strictly smaller rk & deg. $\square$